HERIOT WATT UNIVERSITY

SCHOOL OF MATHEMATICAL AND COMPUTER SCIENCES

Department of Actuarial Mathematics and Statistics

F70LA

Life Insurance Mathematics A

Semester 1 2016/17

Duration: Two Hours

Please answer all 6 questions.

Total marks: 60

Electronic calculators may be used, provided that they conform to University regulations.

You should have a copy of the yellow Formulae and Tables for Actuarial Examinations on your desk.

Question 1

Using the mortality table PFA92C20, calculate the following:

(a) p_{78} [1 mark]

(b) ${}_4q_{63}$ [1 mark]

(c) The probability that a life aged 70 will die between ages 80 and 90. [2 marks]

(d) ${}_{2.7}p_{77}$,assuming uniform distribution of deaths between integer ages [2 marks]

(e) $\mu_{80.7}$, assuming that the force of mortality is constant between integer ages. [2 marks]

[Total 8 marks]

Question 2

For English Life Tables 15 (Males) and (Females), compare the trend in q_x between ages 12 and 29. Suggest possible explanations for your observations.

[Total 7 marks]

Question 3

A life office sells a four-year term assurance contract to a life aged 52 exact. The sum assured is payable immediately on death. The sum assured is calculated as £(400,000- 100,000t) on death at time t years.

Calculate an approximation for the single premium, charged at outset.

You should use Simpson's Rule, with a step size of 2 years. The basis is:

Mortality: Gompertz' Law, with B = 0.0005 and c = 1.07.

Interest: 25% per year

Expenses: none [Total 10 marks]

Question 4

On 1 January 2005, a life office sold a deferred-annuity contract to a man then aged 40, in exchange for a single premium.

The contract said that if the man survives to age 60, he would receive an annuity of £10,000 per year payable continuously from 1 January 2025.

The annuity has a five-year guarantee: if the man dies between ages 60 and 65, the annuity will continue until 1 January 2030.

If the man dies before age 60, the single premium will be refunded to him immediately.

The premium basis and the valuation basis are as follows:

Mortality: AM92 select

Interest: 4% per year effective

Expenses: initial expenses of 10% of the premium; and renewal expenses of 0.5% of the annuity, paid continuously.

(a) Calculate the single premium. [4 marks]

Suppose that on 1 January 2015, there were 200 policies in force, all issued on 1 January 2005 to men then aged 40.

Suppose also that in 2015 there was one death, on 1 April 2015, and interest was earned at a rate of 2.1% per year effective.

(b) Calculate the profit (or loss) to the life office in 2015. Explain why there was a profit (or loss). [7 marks]

[Total 11 marks]

Question 5

On 1 January 2008, an insurance company issued a 28-year with-profit endowment assurance policy to a man then aged 32 exact. The sum assured was £100,000.

The sum assured is payable immediately on death, or on survival to the end of the term. Premiums are payable quarterly in advance throughout the term, or until earlier death.

The first draft of the pricing basis was as follows:

Mortality: AM92 Select

Interest: 4% per year effective

Bonus: Simple, 5% per year

Initial expenses: £1,000

Renewal expenses: 3% of each premium, including the first.

(a) Calculate the quarterly premium, using this first draft as the basis. [7 marks]

The life office eventually decides that it will charge a quarterly premium of £1,350. At 31 December 2014, the total bonuses already declared amount to £43,500.

The gross premium reserving basis is as follows:

Mortality: AM92 Ultimate, rated up by five years i.e. assuming that the life is 5 years older than he actually is.

Interest: 6% per year effective

Bonus: Compound, 1.923% per year

Expenses: 6% of each premium

(b) Calculate the reserve per policy as at 31 December 2014. [5 marks]

(c) A student comments that the reserve seems very low compared to the total premiums that have been paid. Comment on this suggestion. [4 marks]

[Total 16 marks]

Question 6

A life office prices whole of life contracts, with a benefit payable immediately on death. They use a standard mortality table, μ_{x+t} , and a constant force of interest, δ. Expenses can be ignored.

A certain non-standard life, aged x, is considered to carry an extra risk. His force of mortality at any future time t is estimated as $(\mu_{x+t} + \varphi)$, where φ is a constant.

Obtain a simplified expression for the function $\bar{A}_x$ for this non-standard life, in terms of functions calculated on the standard mortality table.

[Total 8 marks]

END OF PAPER